

Python Development
Avni goel | TEN/PY/1206
assignment code 3 (day 3)

Question – Number Pattern Generator

Code –

```
print("Enter your choice :")
print("1. Right triangle of stars")
print("2. Inverted triangle of numbers")
print("3. pascal's triangle")
print("4. Prime numbers upto n")
choice = int(input())
if choice == 1:
    n = int(input("Enter the number of rows : "))
    for i in range(1, n + 1):
        print("* " * i)
elif choice == 2:
    n = int(input("Enter the number of rows : "))
    for i in range(n, 0, -1):
        print(" ".join(str(x) for x in range(1, i + 1)))
elif choice == 3:
    n = int(input("Enter the number of rows : "))
    triangle = []
    for i in range(n):
        row = []
```

```

for j in range(i + 1):
    if j == 0 or j == i:
        row.append(1)
    else:
        value = triangle[i-1][j-1] + triangle[i-1][j]
        row.append(value)
    triangle.append(row)
for row in triangle:
    for num in row:
        print(num, end=" ")
    print()
elif choice == 4:
    n = int(input("Enter the number n : "))
    print("Prime numbers up to", n, ":")
    for num in range(2, n + 1):
        is_prime = True
        for i in range(2, int(num**0.5) + 1):
            if num % i == 0:
                is_prime = False
                break
        if is_prime:
            print(num, end=" ")
else:
    print("Invalid choice!")

```

```
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choice = int(input())
if choice == 1:
    n = int(input("Enter the number of rows : "))
    for i in range(1, n + 1):
        print("* " * i)
elif choice == 2:
    n = int(input("Enter the number of rows : "))
    for i in range(n, 0, -1):
        print(" ".join(str(x) for x in range(1, i + 1)))
elif choice == 3:
    n = int(input("Enter the number of rows : "))
    triangle = []
    for i in range(n):
        row = []
        for j in range(i + 1):
            if j == 0 or j == i:
                row.append(1)
            else:
                value = triangle[i-1][j-1] + triangle[i-1][j]
                row.append(value)
        triangle.append(row)
    for row in triangle:
        for num in row:
            print(num, end=" ")
        print()
```

```

elif choice == 4:
    n = int(input("Enter the number n : "))
    print("Prime numbers up to", n, ":")
    for num in range(2, n + 1):
        is_prime = True
        for i in range(2, int(num**0.5) + 1):
            if num % i == 0:
                is_prime = False
                break
        if is_prime:
            print(num, end=" ")
    else:
        print("Invalid choice!")

```

Output –

```

Enter your choice :
1. Right triangle of stars
2. Inverted triangle of numbers
3. pascal's triangle
4. Prime numbers upto n
1
Enter the number of rows : 5
*
* *
* * *
* * * *
* * * * *

```

```
Enter your choice :
1. Right triangle of stars
2. Inverted triangle of numbers
3. pascal's triangle
4. Prime numbers upto n
2
Enter the number of rows : 5
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

```
Enter your choice :
1. Right triangle of stars
2. Inverted triangle of numbers
3. pascal's triangle
4. Prime numbers upto n
3
Enter the number of rows : 6
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
```

```
Enter your choice :
1. Right triangle of stars
2. Inverted triangle of numbers
3. pascal's triangle
4. Prime numbers upto n
4
Enter the number n : 13
Prime numbers up to 13 :
2 3 5 7 11 13
```